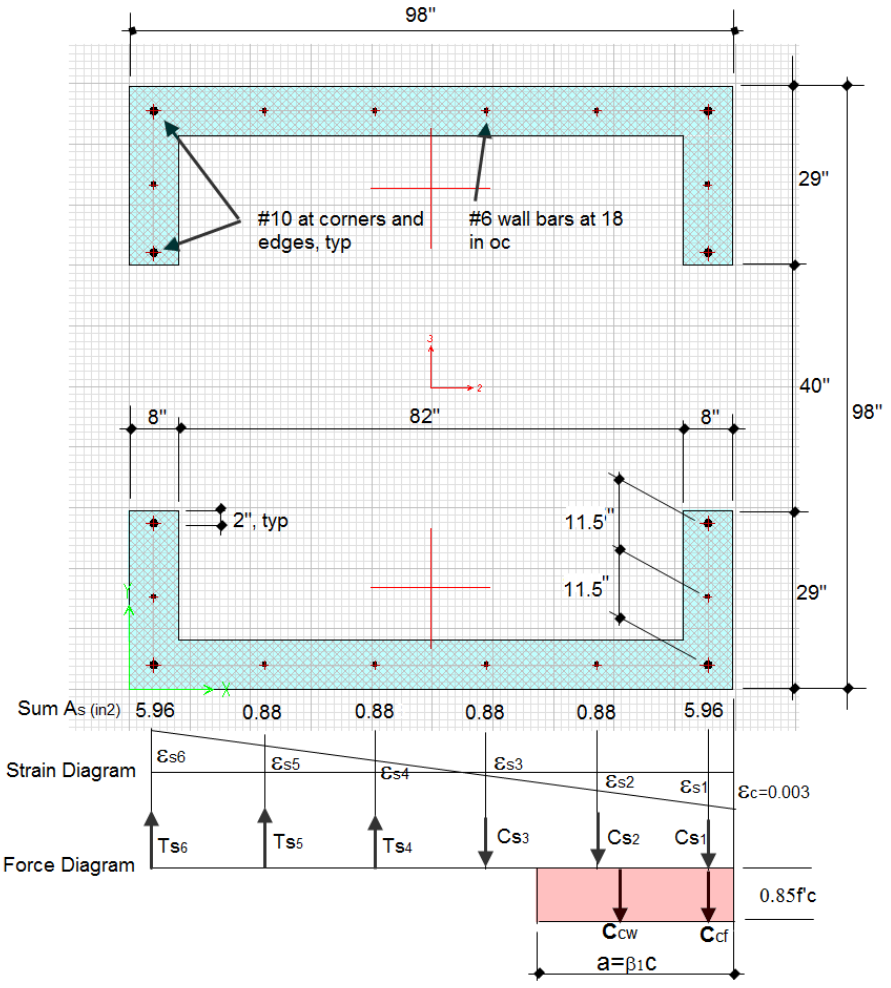
EXAMPLE ACI 318-19 Wall-002

P-M INTERACTION CHECK FOR WALL

EXAMPLE DESCRIPTION

The Demand/Capacity ratio for a given axial loading and moment is tested in this example. The wall is reinforced as shown below. The concrete core wall is loaded with a factored axial load P_u = 2384 k and moments M_{u3} = 9293k-ft. The design capacity ratio is checked by hand calculations and results are compared with ETABS program results.

GEOMETRY, PROPERTIES AND LOADING



Material Properties		Section Properties	Design Properties
E =	3600 k/in^2	tb = 8 in	$f'_c = 4$ k/in^2
v =	0.2	h = 98 in	$f_y = 60$ k/in^2
G =	1500 k/in^2	As1= As6 = 2-#10,2#6 (5.96 in^2)	
		As2, As3, As4 and As5 = 2-#6 (0.88 in^2)	

TECHNICAL FEATURES OF ETABS TESTED

- Concrete wall flexural Demand/Capacity ratio

RESULTS COMPARISON

Independent results are hand calculated and compared with ETABS design check.

Output Parameter	ETABS	Independent	Percent Difference
Wall Demand/Capacity Ratio	1.002	1.003	0.10%

COMPUTER FILE: ACI 318-19 WALL-002

CONCLUSION

The ETABS results show an acceptable comparison with the independent results.

HAND CALCULATION

Wall Strength under compression and bending

1) A value of $e = 46.78$ inches was determined using $e = M_u / P_u$ where M_u and P_u were taken from the ETABS test model interaction diagram. The values of M_u and P_u were large enough to produce a flexural D/C ratio very close to or equal to one. The depth to the neutral axis, *c*, was determined by iteration using an excel spreadsheet so that equations 1 and 2 below were equal.

2) From the equation of equilibrium:

$$P_{n1} = C_c + C_s - T$$

where

$C_c = C_{cw} + C_{cf}$, where C_{cw} and C_{cf} are the area of the concrete web and flange in compression

$$C_{cw} = 0.85 f_c' \bullet 8 \bullet (a - 8)$$

$$C_{cf} = 0.85 f_c' \left(8 \bullet (98 - 40)\right)$$

$$C_s = A_1' \left(f_{s1} - 0.85 f_c'\right) + A_2' \left(f_{s2} - 0.85 f_c'\right) + A_3' \left(f_{s3} - 0.85 f_c'\right)$$

$$T = A_{s4} f_{s4} + A_{s5} f_{s5} + A_{s6} f_{s6}$$

$$P_{n1} = 0.85 f_c' \bullet 8 \bullet (a - 8) + 0.85 f_c' \left(8 \bullet (98 - 40)\right) + A_1' \left(f_{s1} - 0.85 f_c'\right) + A_2' \left(f_{s2} - 0.85 f_c'\right) + A_3' \left(f_{s3} - 0.85 f_c'\right) + A_{s4} f_{s4} + A_{s5} f_{s5} + A_{s6} f_{s6} \quad \text{(Eqn. 1)}$$

3) Taking moments about A_{s6}:

$$P_{n2} = \frac{1}{e'} \left[\begin{array}{l} C_{cf} \left(d - d'\right) + C_{cw} \left(d - \frac{a - t_f}{2} - t_f \right) + C_{s1} \left(d - d'\right) + \\ C_{s2} \left(4s\right) + C_{s3} \left(3s\right) - T_{s4} \left(2s\right) - T_{s5} \left(s\right) \end{array} \right] \quad \text{(Eqn. 2)}$$

where $C_{s1} = A_1' \left(f_{s1} - 0.85 f_c'\right)$, $C_{sn} = A_n' \left(f_{sn} - 0.85 f_c'\right)$, $T_{sn} = f_{sn} A_{sn}$, and the bar strains are determined below. The plastic centroid is at the center of the section, d''

$$= \frac{98 - 8}{2} = 45 \text{ inches}$$

$$e' = e + d'' = 46.78 + 45 = 91.78 \text{ inches}$$

4) Iterating on a value of c until equations 1 and 2 are equal c is found to be $c = 44.58$ inches.

$a = 0.85 \bullet c = 0.85 \bullet 44.58 = 37.89$ inches

5) Assuming the extreme fiber strain equals 0.003 and $c = 44.58$ inches, the steel stresses and strains can be calculated. When the bar strain exceeds the yield strain then, $f_s = f_y$:

$$\varepsilon_{s1} = \left(\frac{c-d'}{c}\right)0.003 = 0.00273; f_s = \varepsilon_s E \le F_y \ ; \ f_{s1} = 60.00 \text{ ksi}$$

$$\varepsilon_{s2} = \left(\frac{c-s-d'}{c}\right)0.003 = 0.00152 \qquad f_{s2} = 44.07 \text{ ksi}$$

$$\varepsilon_{s3} = \left(\frac{c-2s-d'}{c}\right)0.003 = 0.00310 \qquad f_{s3} = 8.94 \text{ ksi}$$

$$\varepsilon_{s4} = \left(\frac{d-c-2s}{d-c}\right)\varepsilon_{s6} = 0.00090 \qquad f_{s4} = 26.2 \text{ ksi}$$

$$\varepsilon_{s5} = \left(\frac{d-c-s}{d-c}\right)\varepsilon_{s6} = 0.00211 \qquad f_{s5} = 60.00 \text{ ksi}$$

$$\varepsilon_{s6} = \left(\frac{d-c}{c}\right)0.003 = 0.00333 \qquad f_{s6} = 60.00 \text{ ksi}$$

Substituting the above values of the compression block depth, a, and the rebar stresses into equations Eqn. 1 and Eqn. 2 give

$P_{n1} = 3148$ k
$P_{n2} = 3148$ k

$M_n = P_n e = 3148(46.78)/12 = 12{,}273$ k-ft

6) Determine ϕ based on maximum strain value .

$\varepsilon_t = 0.00333$, $\varepsilon_{ty} = \dfrac{60{,}000}{29{,}000{,}000} = 0.0020689$

For $\varepsilon_{ty} < \varepsilon_t < \varepsilon_{ty} + 0.003$:

$$\phi = \phi_c + (\phi_t - \phi_c)\left(\frac{\varepsilon_t - \varepsilon_{ty}}{0.003}\right) = 0.65 + (0.90 - 0.65)\left(\frac{0.00333 - 0.0020689}{0.003}\right) = 0.755$$

7) Calculate ϕP_n and ϕM_n

$$\phi P_n = 0.755(3148) = 2377 \text{ kips}$$
$$\phi M_n = 0.755(12,273) = 9,266 \text{ k-ft.}$$

8) Calculate the D/C ratio.

$$\text{D/C} = \frac{P_u}{\phi P_n} = \frac{2384}{2377} = 1.003$$
$$\text{D/C} = \frac{M_u}{\phi M_n} = \frac{9293}{9266} = 1.003$$